

Текстовая диффузия

Марк Блуменау

Почему трансформер нам не нравится?

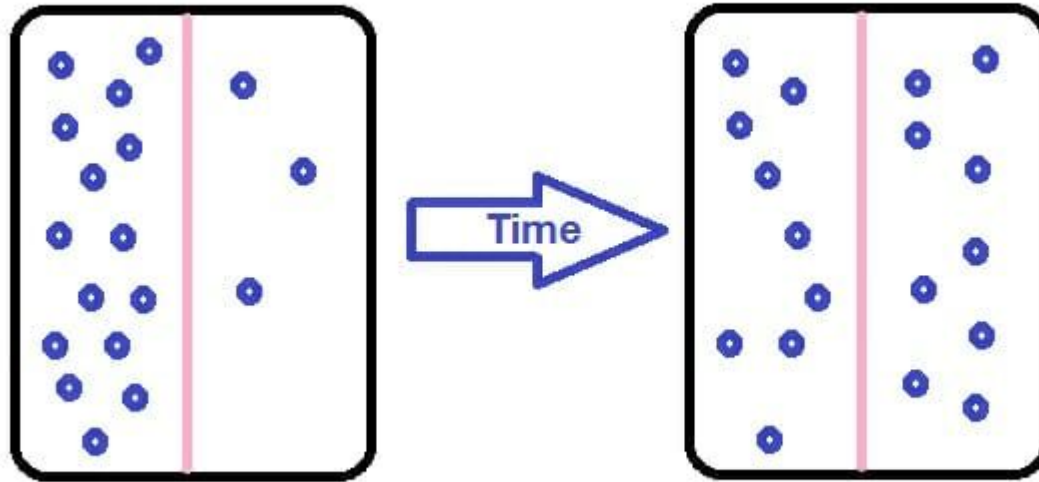
- 1) Не может исправлять ошибки текста после генерации символа
- 2) Дорогой и долгий
- 3) Не всегда естественный (дорогой чатжбт только расставь ссылки)

База



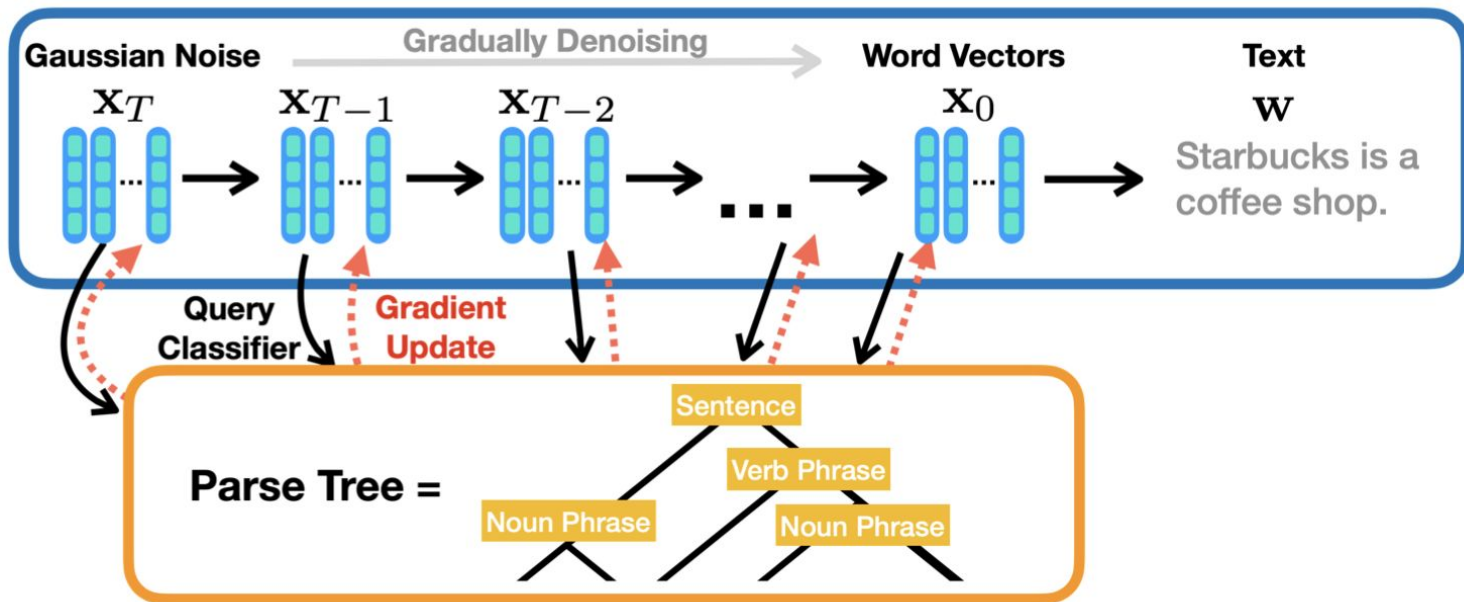
[Why does ChatGPT sometimes contradict itself within the same sentence? : r/ChatGPT](#)

Решение! Давайте делать диффузию

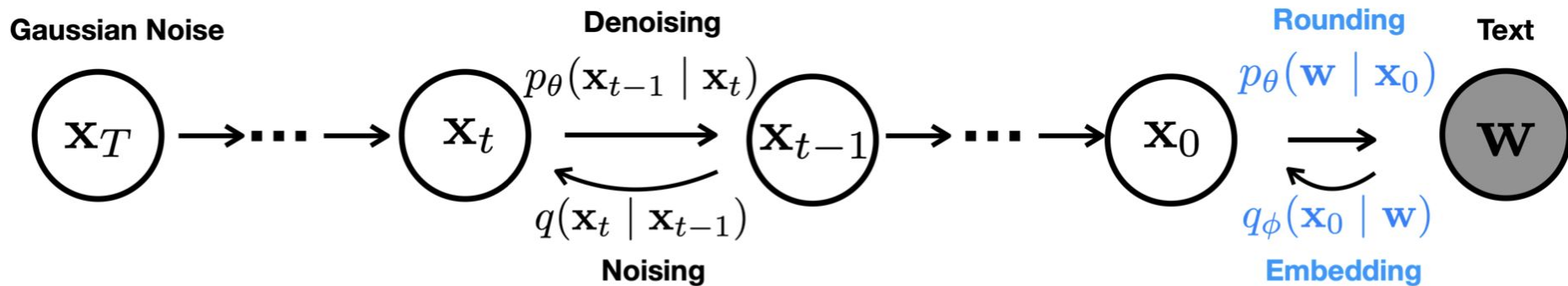


А как делать? Diffusion LM

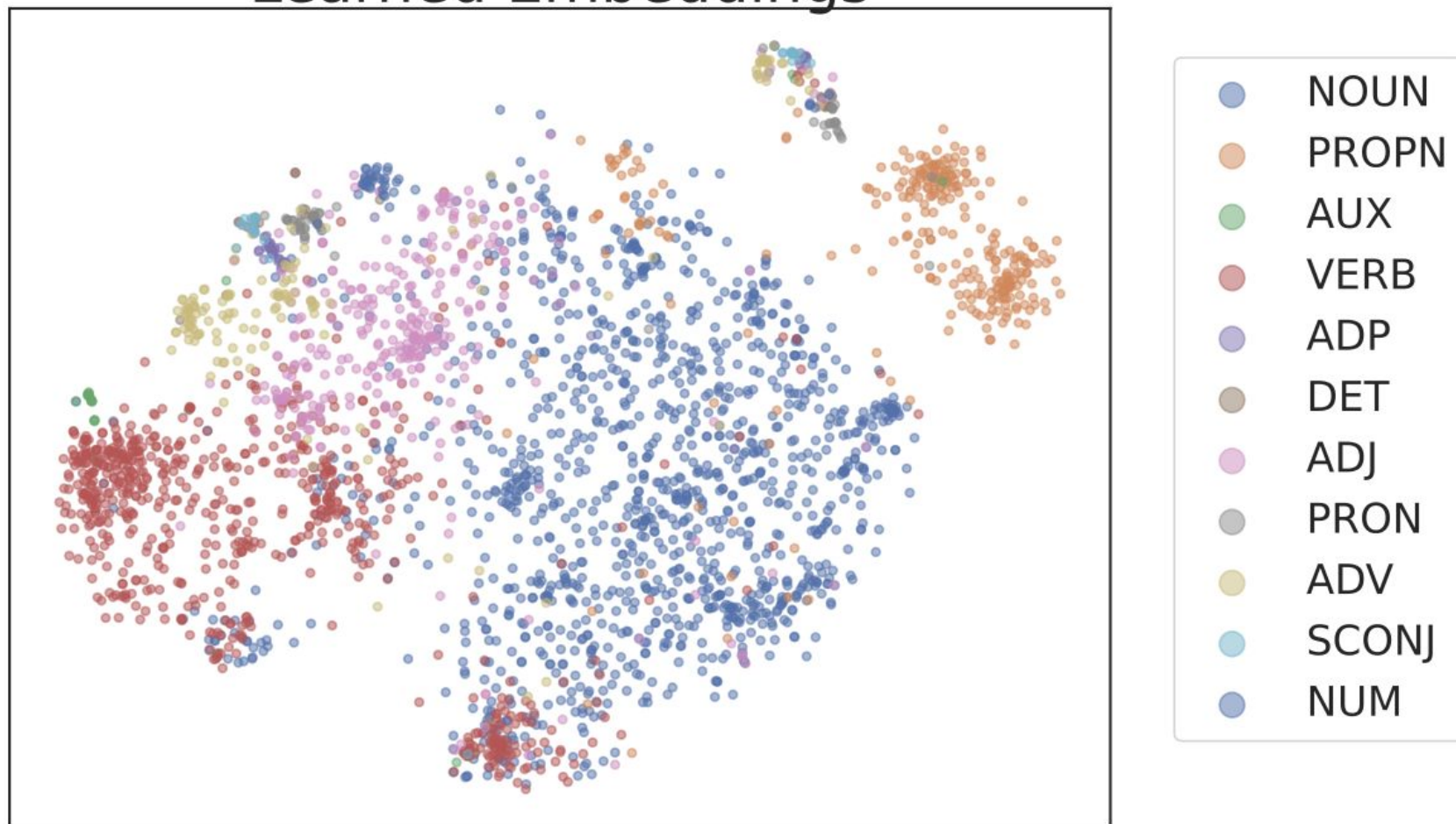
Diffusion-LM



Есть нюанс...



Learned Embeddings



Вот и все, приехали

The learned embeddings define a mapping from discrete text to the continuous $\mathbf{x}_0$. We now describe the inverse process of rounding a predicted $\mathbf{x}_0$ back to discrete text. Rounding is achieved by choosing the most probable word for each position, according to $\operatorname{argmax} p_{\theta}(\mathbf{w} \mid \mathbf{x}_0) = \prod_{i=1}^n p_{\theta}(w_i \mid x_i)$. Ideally, this argmax -rounding would be sufficient to map back to discrete text, as the denoising steps should ensure that $\mathbf{x}_0$ lies exactly on the embedding of some word. However, empirically, the model fails to generate $\mathbf{x}_0$ that commits to a single word.

А давайте спонугаем Stable diffusion

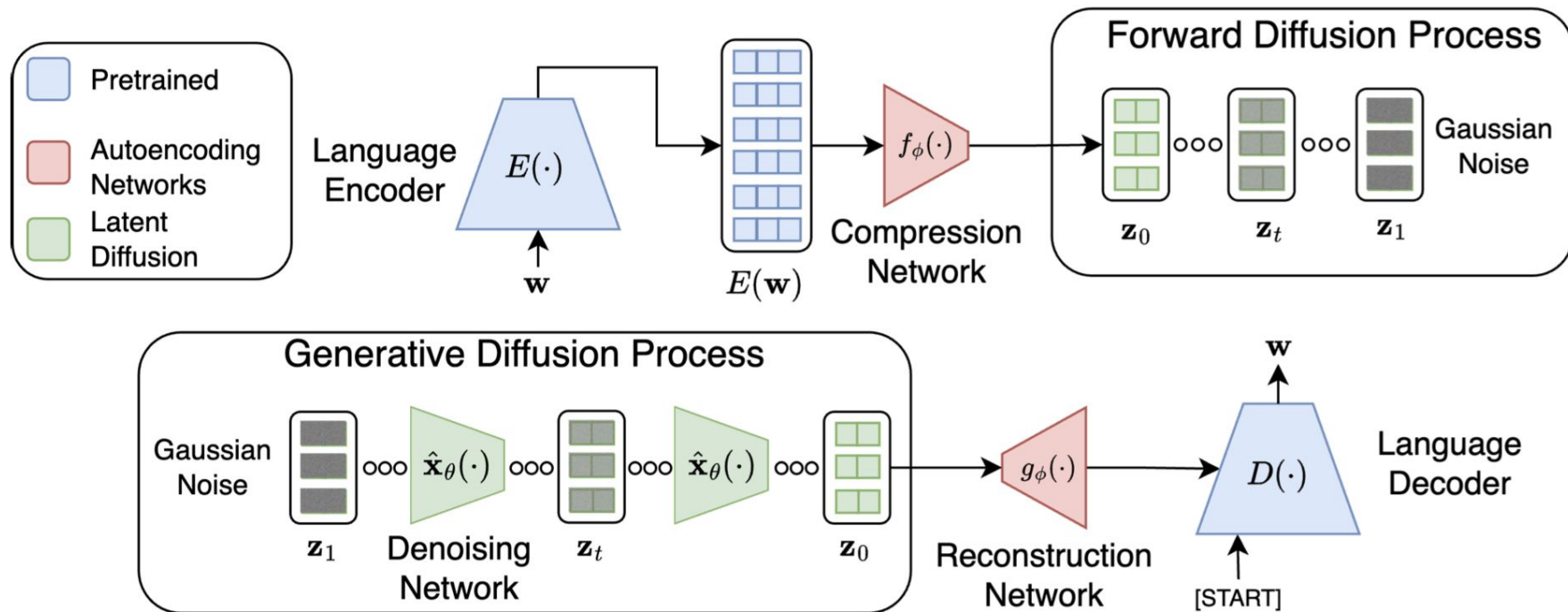


Figure 1: Overview of our proposed latent language diffusion framework.

We use **MAUVE Score** [47] and **Perplexity** (Ppl) to evaluate the quality of our generated text. MAUVE Score is a metric for open-ended text generation that compares the distribution of generated text with that of reference text using divergence frontiers. We follow Pillutla et al. [47] and use the GPT-2-Large model [49] to embed the text. Perplexity measures how likely the generated samples are according to an autoregressive language model; we use GPT-2-Large to compute perplexity.

We also want to quantify the **Diversity** (Div) of generations. We define diversity as $\text{Div} = \prod_{n=2}^4 \frac{|\text{unique } n\text{-grams}(\{\mathbf{w}_i\})|}{|\text{total } n\text{-grams}(\{\mathbf{w}_i\})|}$ where $\{\mathbf{w}_i\}$ is a set of generated samples [68]. The metrics discussed so far can be optimized by generating samples from the training set. We measure the proportion of generated 4-grams that are found in the training set to quantify the degree of **Memorization** (Mem).

IIIIIIII

Table 1

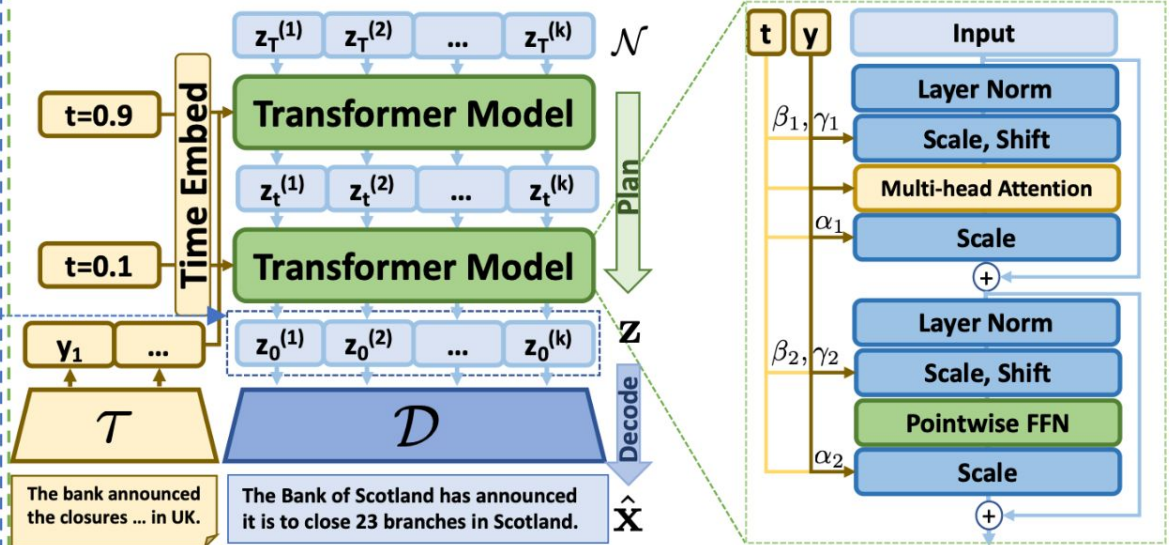
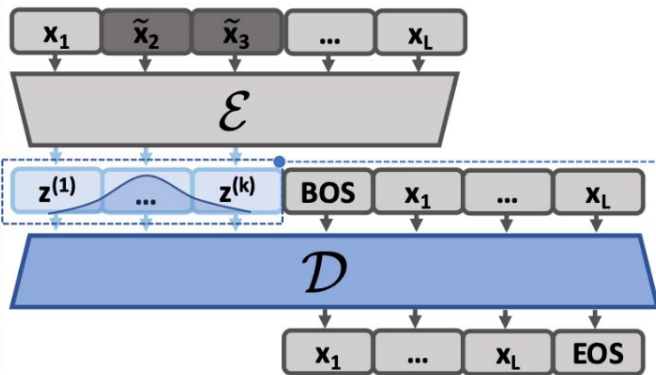
	Timesteps	ROCStories				AG News			
		MAUVE $\uparrow$	Ppl $\downarrow$	Div $\uparrow$	Mem $\downarrow$	MAUVE $\uparrow$	Ppl $\downarrow$	Div $\uparrow$	Mem $\downarrow$
Reference	-	.951 _{.007}	21.1 _{.3}	.414 _{.003}	.362 _{.003}	.951 _{.014}	43.6 _{1.2}	.658 _{.002}	.385 _{.005}
Diffusion-LM [36]	2000	.043 _{.006}	47.3 _{.6}	.128 _{.002}	.434 _{.002}	.012 _{.001}	67.1 _{1.2}	.043 _{.002}	.086 _{.006}
LD4LG (BART-Base)	250	.716 _{.019}	30.6 _{.5}	.331 _{.005}	.441 _{.004}	.866 _{.016}	100.6 _{2.9}	.540 _{.006}	.293 _{.001}
LD4LG (FLAN-T5-base)	250	.481 _{.007}	37.5 _{.4}	.389 _{.002}	.387 _{.002}	.859 _{.020}	122.0 _{3.9}	.624 _{.008}	.221 _{.003}
GPT-2-Medium	-	.788 _{.025}	20.0 _{.2}	.372 _{.002}	.688 _{.006}	.820 _{.012}	37.3 _{1.1}	.532 _{.017}	.829 _{.005}

И чего?

Table 5: Metrics for class-conditional generation.

		LD4LG (BART-base)					LD4LG (FLAN-T5-base)				
		MAUVE ↑				Mem ↓	MAUVE ↑				Mem ↓
Conditioning		World	Sports	Business	Sci/Tech		World	Sports	Business	Sci/Tech	
Diffusion	World	.842 _{.017}	.015 _{.002}	.026 _{.002}	.020 _{.002}	.296 _{.002}	.809 _{.024}	.013 _{.001}	.025 _{.002}	.022 _{.002}	.233 _{.005}
	Sports	.013 _{.001}	.845 _{.024}	.011 _{.001}	.010 _{.000}	.305 _{.003}	.011 _{.001}	.836 _{.020}	.009 _{.000}	.009 _{.000}	.249 _{.004}
	Business	.024 _{.002}	.011 _{.001}	.752 _{.030}	.068 _{.005}	.363 _{.009}	.025 _{.003}	.011 _{.001}	.765 _{.016}	.076 _{.008}	.244 _{.004}
	Sci/Tech	.023 _{.002}	.012 _{.001}	.082 _{.008}	.813 _{.028}	.225 _{.004}	.024 _{.001}	.011 _{.001}	.082 _{.010}	.843 _{.033}	.169 _{.004}
		Conditional GPT-2					Reference				
		MAUVE ↑				Mem ↓	MAUVE ↑				Mem ↓
Conditioning		World	Sports	Business	Sci/Tech		World	Sports	Business	Sci/Tech	
Comparisons	World	.805 _{.022}	.012 _{.000}	.025 _{.002}	.021 _{.002}	.402 _{.002}	.963 _{.009}	.018 _{.001}	.034 _{.002}	.032 _{.003}	.388 _{.007}
	Sports	.017 _{.001}	.840 _{.019}	.012 _{.001}	.013 _{.001}	.369 _{.004}	.018 _{.001}	.958 _{.007}	.014 _{.001}	.014 _{.002}	.346 _{.002}
	Business	.037 _{.003}	.012 _{.001}	.629 _{.029}	.069 _{.007}	.479 _{.007}	.040 _{.005}	.014 _{.001}	.968 _{.009}	.125 _{.009}	.441 _{.003}
	Sci/Tech	.033 _{.002}	.013 _{.001}	.102 _{.015}	.697 _{.027}	.434 _{.004}	.036 _{.003}	.016 _{.001}	.133 _{.013}	.955 _{.011}	.366 _{.003}

А что хитрее?



Fun fact: люди думают Apple не занимается
нейронками

Yizhe Zhang, Jiatao Gu, Zhuofeng Wu, Shuangfei Zhai, Josh Susskind, Navdeep Jaitly
Apple
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Пора компенсировать

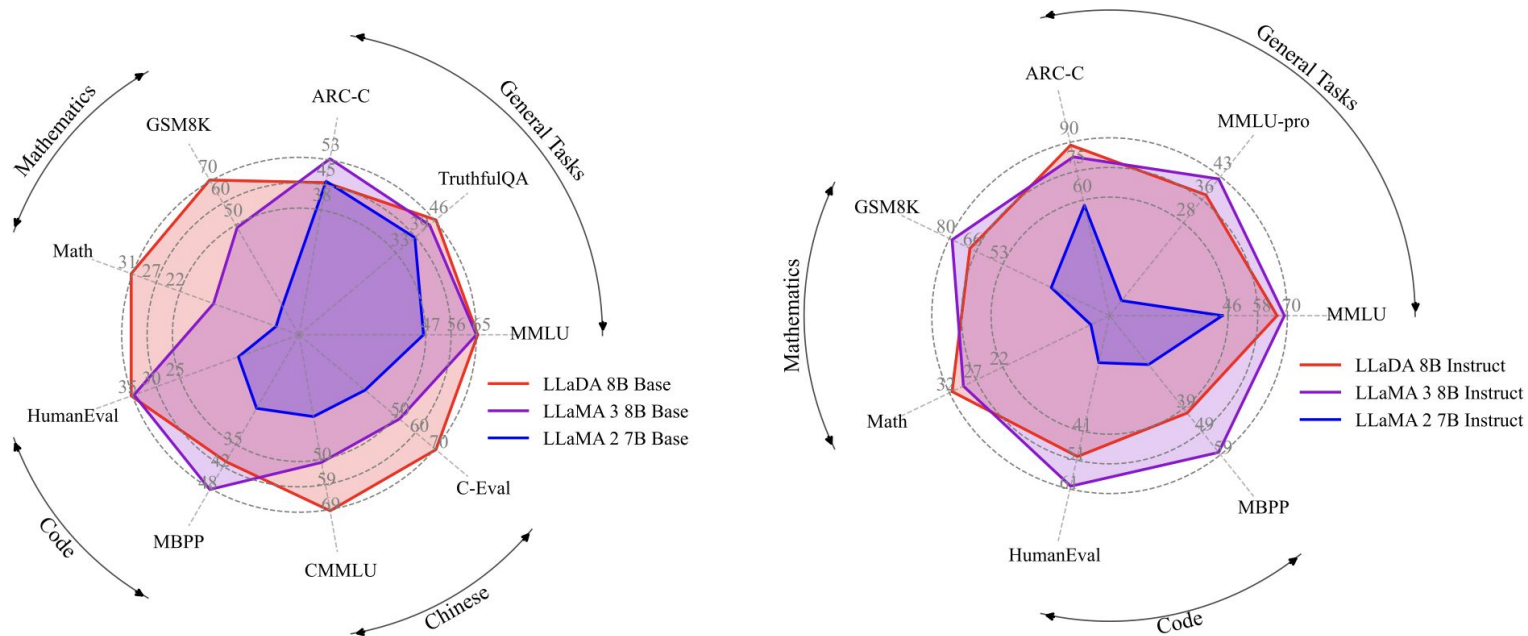


Figure 1: **Zero/Few-Shot Benchmarks.** We scale LLaDA to 8B parameters from scratch and observe competitive zero/few-shot performance compared with strong autoregressive LLMs [6].

Как будет работать компенсация?

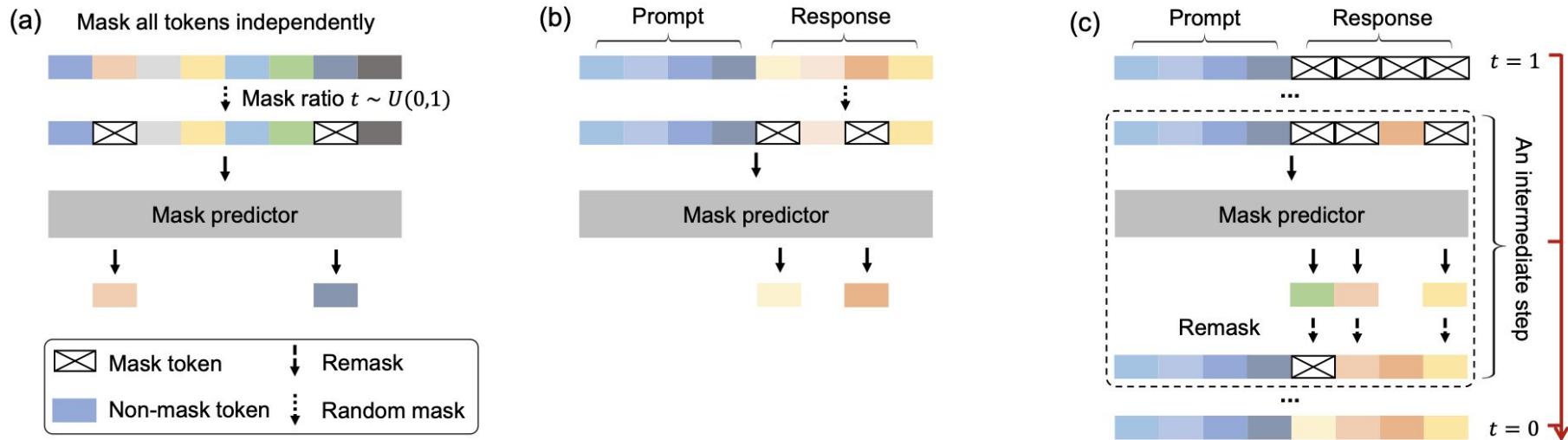
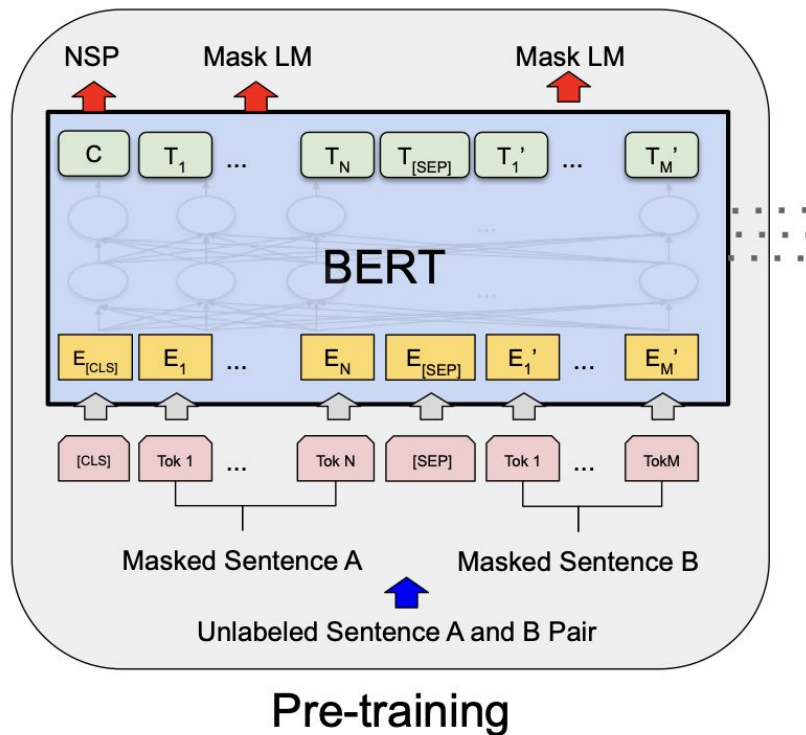


Figure 2: Overview of LLaDA. (a) Pre-training. LLaDA is trained on text with random masks applied independently to all tokens at the same ratio $t \sim U[0, 1]$. (b) SFT. Only response tokens are possibly masked. (c) Sampling. LLaDA simulates a diffusion process from $t = 1$ (fully masked) to $t = 0$ (unmasked), predicting all masks simultaneously at each step with flexible remask strategies.

Стоп, они реально взяли просто берт?



Basically да...

tokens. We refer to this procedure as a “masked LM” (MLM), although it is often referred to as a *Cloze* task in the literature (Taylor, 1953). In this case, the final hidden vectors corresponding to the mask tokens are fed into an output softmax over the vocabulary, as in a standard LM. In all of our experiments, we mask 15% of all WordPiece tokens in each sequence at random. In contrast to denoising auto-encoders (Vincent et al., 2008), we only predict the masked words rather than reconstructing the entire input.

Although this allows us to obtain a bidirectional pre-trained model, a downside is that we are creating a mismatch between pre-training and fine-tuning, since the [MASK] token does not appear during fine-tuning. To mitigate this, we do not always replace “masked” words with the actual [MASK] token. The training data generator chooses 15% of the token positions at random for prediction. If the i -th token is chosen, we replace

Результаты

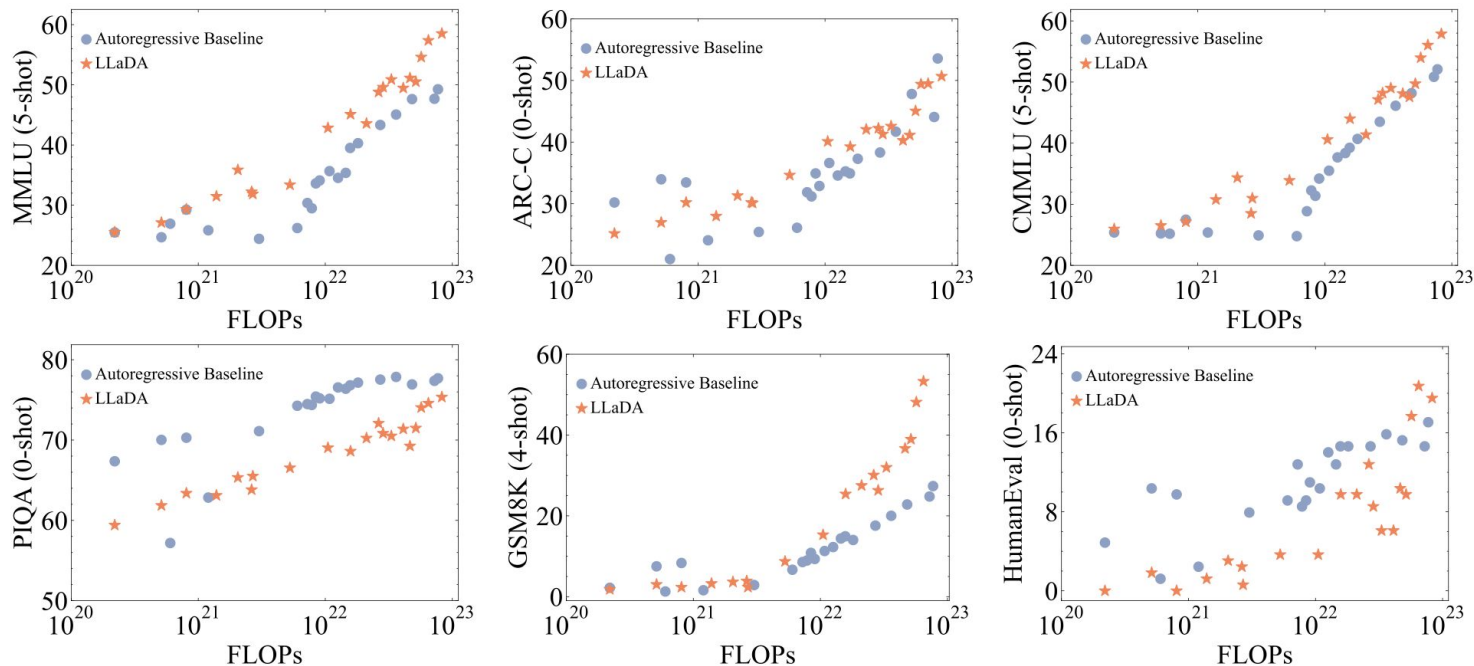


Figure 3: Scalability of LLaDA. We evaluate the performance of LLaDA and our ARM baselines trained on the same data across increasing pre-training computational FLOPs. LLaDA exhibits strong scalability, matching the overall performance of ARMs on six tasks.

Еще метрики

	LLaDA 8B*	LLaMA3 8B*	LLaMA2 7B*	Qwen2 7B [†]	Qwen2.5 7B [†]	Mistral 7B [†]	Deepseek 7B [¶]
Model Training tokens	Diffusion 2.3T	AR 15T	AR 2T	AR 7T	AR 18T	AR -	AR 2T
General Tasks							
MMLU	65.9 (5)	65.4 (5)	45.9 (5)	70.3 (5)	74.2 (5)	64.2 (5)	48.2 (5)
BBH	49.7 (3)	62.1 (3)	39.4 (3)	62.3 (3)	70.4 (3)	56.1 (3)	39.5 (3)
ARC-C	45.9 (0)	53.1 (0)	46.3 (0)	60.6 (25)	63.7 (25)	60.0 (25)	48.1 (0)
Hellaswag	70.5 (0)	79.1 (0)	76.0 (0)	80.7 (10)	80.2 (10)	83.3 (10)	75.4 (0)
TruthfulQA	46.1 (0)	44.0 (0)	39.0 (0)	54.2 (0)	56.4 (0)	42.2 (0)	-
WinoGrande	74.8 (5)	77.3 (5)	72.5 (5)	77.0 (5)	75.9 (5)	78.4 (5)	70.5 (0)
PIQA	73.6 (0)	80.6 (0)	79.1 (0)	-	-	-	79.2 (0)
Mathematics & Science							
GSM8K	70.3 (4)	48.7 (4)	13.1 (4)	80.2 (4)	85.4 (4)	36.2 (4)	17.4 (8)
Math	31.4 (4)	16.0 (4)	4.3 (4)	43.5 (4)	49.8 (4)	10.2 (4)	6.0 (4)
GPQA	25.2 (5)	25.9 (5)	25.7 (5)	30.8 (5)	36.4 (5)	24.7 (5)	-
Code							
HumanEval	35.4 (0)	34.8 (0)	12.8 (0)	51.2 (0)	57.9 (0)	29.3 (0)	26.2 (0)
HumanEval-FIM	73.8 (2)	73.3 (2)	26.9 (2)	-	-	-	-
MBPP	40.0 (4)	48.8 (4)	23.2 (4)	64.2 (0)	74.9 (0)	51.1 (0)	39.0 (3)
Chinese							
CMMLU	69.9 (5)	50.7 (5)	32.5 (5)	83.9 (5)	-	-	47.2 (5)
C-Eval	70.5 (5)	51.7 (5)	34.0 (5)	83.2 (5)	-	-	45.0 (5)

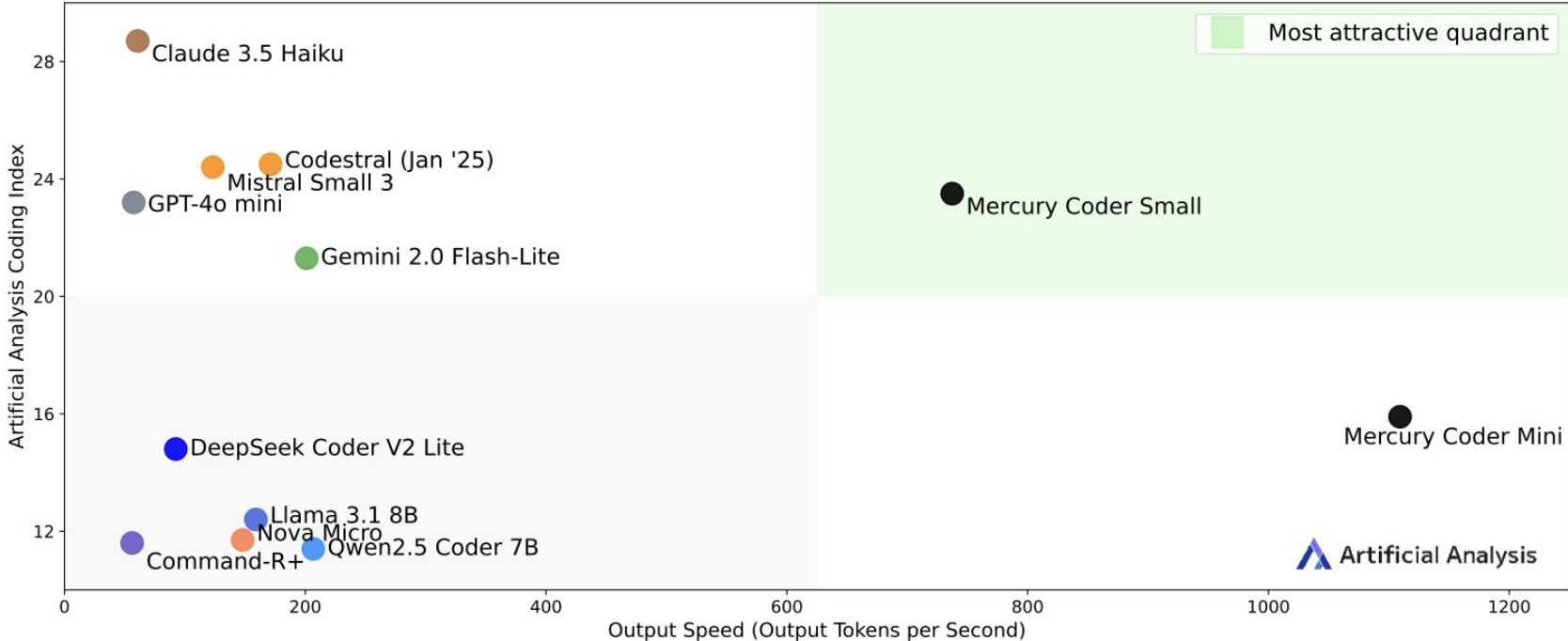
	LLaDA 8B*	LLaMA3 8B*	LLaMA2 7B*	Qwen2 7B [†]	Qwen2.5 7B [†]	Gemma2 9B [†]	Deepseek 7B [¶]
Model	Diffusion	AR	AR	AR	AR	AR	AR
Training tokens	2.3T	15T	2T	7T	18T	8T	2T
Post-training	SFT	SFT+RL	SFT+RL	SFT+RL	SFT+RL	SFT+RL	SFT+RL
Alignment pairs	4.5M	-	-	0.5M + -	1M + 0.15M	-	1.5M + -
General Tasks							
MMLU	65.5 (5)	68.4 (5)	44.1 (5)	-	-	-	49.4 (0)
MMLU-pro	37.0 (0)	41.9 (0)	4.6 (0)	44.1 (5)	56.3 (5)	52.1 (5)	-
Hellaswag	74.6 (0)	75.5 (0)	51.5 (0)	-	-	-	68.5 (-)
ARC-C	88.5 (0)	82.4 (0)	57.3 (0)	-	-	-	49.4 (-)
Mathematics & Science							
GSM8K	69.4 (4)	78.3 (4)	29.0 (4)	85.7 (0)	91.6 (0)	76.7 (0)	63.0 (0)
Math	31.9 (0)	29.6 (0)	3.8 (0)	52.9 (0)	75.5 (0)	44.3 (0)	15.8 (0)
GPQA	33.3 (5)	31.9 (5)	28.4 (5)	34.3 (0)	36.4 (0)	32.8 (0)	-
Code							
HumanEval	49.4 (0)	59.8 (0)	16.5 (0)	79.9 (0)	84.8 (0)	68.9 (0)	48.2 (-)
MBPP	41.0 (4)	57.6 (4)	20.6 (4)	67.2 (0)	79.2 (0)	74.9 (0)	35.2 (-)

Sampling Process	
User	Lily can run 12 kilometers per hour for 4 hours. After that, she runs 6 kilometers per hour. How many kilometers can she run in 8 hours?
LLaDA	In4hours, Lily runs 12 * 4 = 48 kilometers . After 4 hours she runs 6 * 4 = 24 kilometers . In total she runs 48 + 24 = 72 kilometers in 8 hours . Result 72
Multi-round Dialogue Case	
User	Could you please share with me what the initial two lines of the renowned poem ‘The Road Not Taken’?
LLaDA	Certainly! The initial two lines of the renowned poem “The Road Not Taken" by Robert Frost are: “Two roads diverged in a yellow wood, And sorry I could not travel both." These lines set the stage for the poem, introducing the idea of a speaker facing a choice between two paths.
User	Please help me translate into Chinese.
LLaDA	Sure! The first two lines of “The Road Not Taken" by Robert Frost can be translated into Chinese as: “两条路分岔在黄色的树林中，遗憾我不能同时走”
User	Please translate into German.
LLaDA	Sure! The first two lines of “The Road Not Taken" by Robert Frost can be translated into German as: “Zwei Wege trennten sich im gelben Wald, und leider konnte ich nicht beide tun."
User	Please also write a poem about life choices. I require 5 sentences, and the first word of each sentence starts with C as the first letter.
LLaDA	Certainly! Here’s a poem about life choices, with each sentence starting with the letter “C”: Choices cascade through life’s journey, Creating connections with each decision, Careful consideration leads to better paths, Courageous choices shape our destiny, Choosing wisely guides us through the years.

Ртуть? Ртуть.

Coding Index vs. Output Speed: Smaller models

Artificial Analysis Coding Index (represents the average of LiveCodeBench & SciCode);
Output Speed: Output Tokens per Second; 1,000 Input Tokens; Coding focused workload



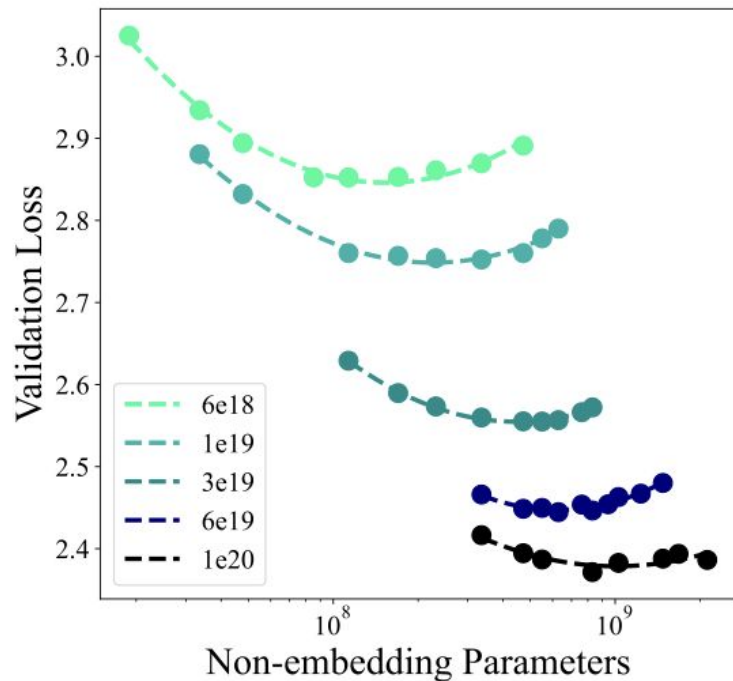
Model	HumanEval*	MBPP	EvalPlus	MultiPL-E	LCB*	BCB	Speed
Open-Weights Models							
Llama 3.1 8B Instruct	66.5	59.2	60.2	50.1	12.0	32.3	153
DeepSeek Coder V2 Lite	79.0	59.8	68.3	57.0	16.0	44.4	93
Mistral Small 3	84.8	69.6	72.3	70.1	25.0	42.7	126
Qwen 2.5 Coder 7B Instruct	88.0	80.0	79.3	75.3	9.0	41.4	195
Frontier Speed-Optimized Models							
Nova Micro	79.3	65.4	72.1	56.7	14.0	-	148
Codestral 2501	85.0	72.2	75.6	73.4	24.0	46.1	171
GPT 4o Mini	88.0	74.6	78.5	72.0	23.0	46.8	59
Claude 3.5 Haiku	86.0	78.0	75.1	72.3	31.0	45.4	61
Gemini 2.0 Flash Lite	90.0	75.0	77.3	79.5	18.0	44.4	201
Frontier Models							
DeepSeek V3	92.1	81.0	82.1	79.1	36.0	50.0	27
Claude 3.5 Sonnet	90.2	81.2	77.3	81.9	38.0	44.8	76
GPT 4o	90.2	82.2	82.4	77.6	31.0	49.9	61
Our Models							
Mercury Coder Mini	88.0	77.1	78.6	74.1	17.0	42.0	1109
Mercury Coder Small	90.0	76.6	80.4	76.2	25.0	45.5	737

Model	CPP	Java	JS	PHP	Bash	TS	Avg
Open-Weights Models							
Llama 3.1 8B Instruct	54.0	48.7	57.8	49.1	34.8	55.9	50.1
OpenCoder 8B Instruct	70.2	70.8	78.9	72.1	44.1	75.2	68.5
Mistral Small 3	74.5	73.9	82.0	66.5	43.5	80.1	70.1
Qwen 2.5 Coder 14B Instruct	77.6	55.9	83.9	61.5	46.0	83.9	68.1
Frontier Speed-Optimized Models							
Nova Micro	59.6	57.1	67.7	62.1	23.6	70.2	56.7
Codestral 2501	80.1	72.7	83.2	73.9	47.2	83.2	73.4
GPT 4o Mini	78.3	73.4	82.0	71.4	46.6	80.1	72.0
Claude 3.5 Haiku	75.2	78.5	79.5	73.3	45.3	82.0	72.3
Gemini 2.0 Flash Lite	84.5	82.6	88.2	85.7	50.9	85.1	79.5
Frontier Models							
DeepSeek V3	84.5	73.9	87.6	81.4	59.0	88.2	79.1
Claude 3.5 Sonnet	82.0	88.6	89.4	83.2	57.8	90.6	81.9
GPT 4o	79.5	81.0	87.0	78.3	52.8	87.0	77.6
Our Models							
Mercury Coder Mini	78.9	74.5	78.9	72.7	56.5	83.2	74.1
Mercury Coder Small	82.0	80.1	83.9	78.3	50.1	82.6	76.2

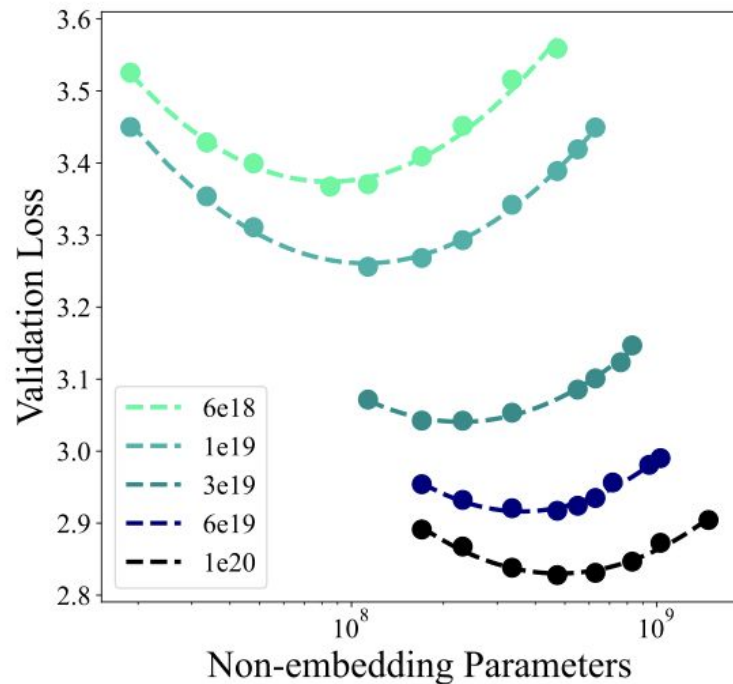
Table 4: Co-Pilot Arena model comparison by latency, Elo scores, and ranks. Data obtained via Copilot Arena.

Model	Latency (seconds)	Latency Rank	Elo Score	Elo Rank
DeepSeek V2.5 (FIM)	2.07	11	1025	1
Claude 3.5 Sonnet	1.46	8	1003	1
Mercury Coder Mini	0.25	1	993	2
Codestral	0.31	2	992	2
Metal Llama 3.1 405B	1.84	10	982	3
GPT-4o	0.76	5	980	3
Gemini 1.5 Flash	0.59	3	977	3
Gemini 1.5 Pro	1.48	9	977	5
Meta Llama 3.1 70B	1.17	7	970	5
Qwen 2.5 Coder 32B	0.91	6	949	12
GPT-4o Mini	0.84	4	939	12

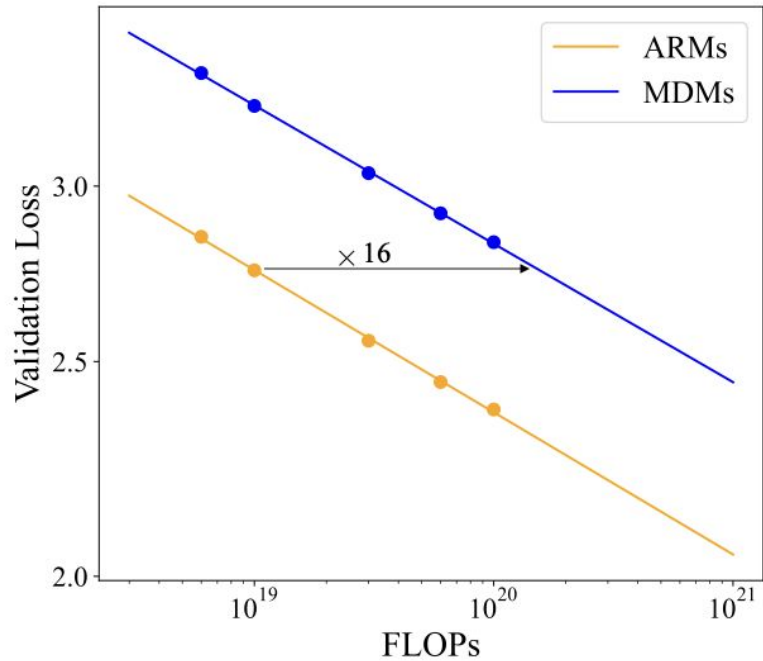
What is the catch?



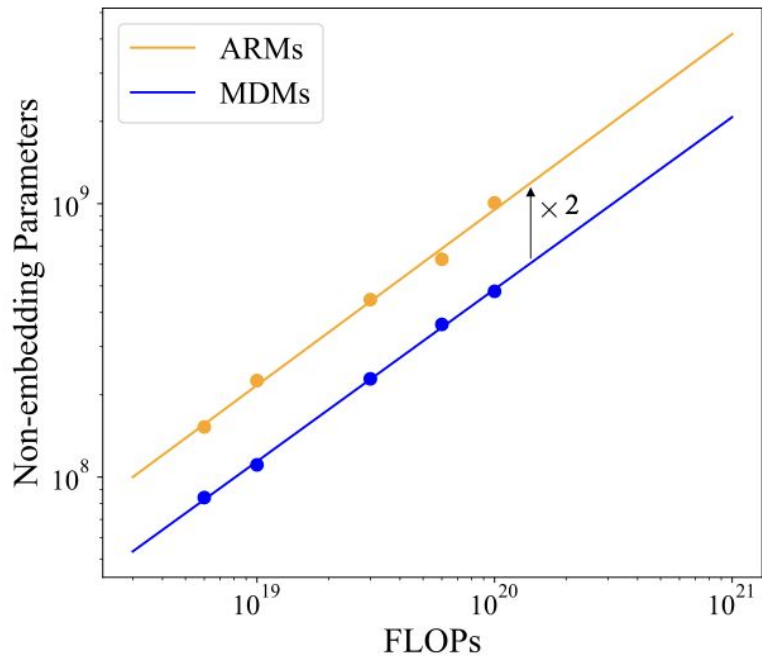
(a) ARMs.



(b) MDMs.



(a) Loss-Flops curve.



(b) Parameters-Flops curve.